

Polynomial-Time Algorithm for Computing K -Terminal Reliability of d -Trapezoid Graphs

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Abstract

Consider a probabilistic graph in which the edges are perfectly reliable, but vertices may fail with some known probabilities. The K -terminal reliability of this graph is the probability that a given set of vertices K are connected to each other. This reliability problem is known to be #P-complete for general graphs, and it remains #P-complete for chordal graphs and comparability graphs. This work proposes an $O(n^{2d+1})$ -time algorithm for computing the K -terminal reliability of d -trapezoid graphs.

polynomial-time algorithms for computing the K -terminal reliability of such interval graphs and permutation graphs.

Our earlier work [5] proposed a linear-time algorithm for computing the K -terminal reliability of proper interval graphs. This paper extends the problem of computing the K -terminal reliability of d -trapezoid graphs, which are common generalizations of interval, permutation and trapezoid graphs, such that 1-trapezoid graphs coincide with interval graphs and 2-trapezoid graphs coincide with trapezoid graphs.

1 Introduction

The network model that is presented herein is a graph with perfect edges and imperfect vertices. The failure probabilities of the vertices are assumed to be mutually independent. A subset of K vertices is chosen as the target vertices of the network. The K -terminal reliability is the probability that all vertices in K are connected to each other by a set of working vertices. This network model can be applied to wireless networks [1,2] in which communication links are perfectly reliable, but in which sites may fail with some known probabilities. AboEIFotoh and Colbourn [3,4] confirmed that this problem is #P-complete for general graphs and it remains so even for chordal graphs and comparability graphs. AboEIFotoh and Colbourn [4] have also proposed

2 Preliminaries

This section begins with some definitions and properties that are related to d -trapezoid graphs.

Definition 1. A graph $G=(V,E)$ is a *trapezoid graph* if there exists a set Q of trapezoids between two parallel lines, such that each vertex $v \in V$ corresponds to a trapezoid $Q_v \in Q$ and an edge $(v,w) \in E$ if and only if trapezoids Q_v and Q_w intersect each other. Each trapezoid Q_v has an interval $[l_v^1, r_v^1]$ on the top line and an interval $[l_v^2, r_v^2]$ on the bottom line. The above geometric representation is called a trapezoid diagram, which is denoted by $D(G)$.

Without loss of generality, all points l_v^1, r_v^1 , for all $v \in V$, on the top line (similarly, l_v^2, r_v^2 on the

bottom line) are labeled with distinct integers between 1 and $2n$, where n is the number of vertices in the trapezoid graph. Flotow [6] generalized the definition of trapezoid graphs as follows.

Definition 2 [6,7]. Let d be a position integer and let $D^1, D^2, \dots, D^d$ be d parallel lines from top to bottom. A graph $G=(V,E)$ is a d -trapezoid graph if there are families of intervals $I^i=\{[l_v^i, r_v^i] \mid v \in V; l_v^i, r_v^i \in \{1, 2, \dots, 2n\}; l_v^i < r_v^i\}$ on D^i for all

$i \in \{1, 2, \dots, d\}$, satisfying $(v, w) \in E$ if and only if Q_v and Q_w intersect each other, where Q_v represents the closed polygon $l_v^1, \dots, l_v^d, r_v^d, \dots, r_v^1$. Given a d -trapezoid graph $G=(V,E)$, the d -trapezoid representation $I^i=\{[l_v^i, r_v^i] \mid v \in V; l_v^i, r_v^i \in D^i; l_v^i < r_v^i\}$ for all $i \in \{1, 2, \dots, d\}$ is called the d -trapezoid diagram of G , which is denoted by $D(G)$. The polygon Q_v is said to be a d -trapezoid in a d -trapezoid diagram. Figure 1 presents a 3-trapezoid graph and its corresponding 3-trapezoid diagram.

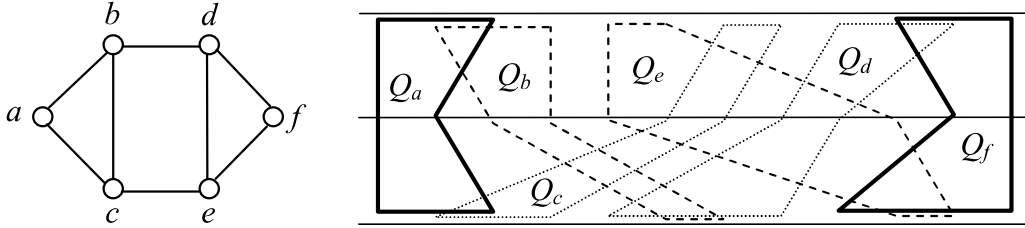


Figure 1: Example of a 3-trapezoid graph G and its corresponding 3-trapezoid diagram $D(G)$.

Theorem 1 [6,7]. The d -trapezoid graphs are identically the cocomparability graphs with interval dimensions of at most d .

The concept of scanlines was introduced elsewhere [7,8], and has been used to solve many problems on d -trapezoid graphs.

Definition 3 [7]. Given one point $s^i \in \{\frac{1}{2}, 1\frac{1}{2}, \dots, 2n\frac{1}{2}\}$ on each line D^i , a scanline s of the d -trapezoid diagram $D(G)$ is the polyline $(s^1, s^2, \dots, s^d)$. Let $Q(s)$ be the set of d -trapezoids Q_v for which s has a non-empty intersection with Q_v in $D(G)$, such that $Q(s) \equiv \{Q_v \mid Q_v \in D(G); Q_v \cap s \neq \emptyset\}$.

Definition 4 [7]. Let s_x and s_y be two scanlines. The scanline s_x is left of s_y , denoted by $s_x \prec s_y$, if $s_x^i \leq s_y^i$ for all $i \in \{1, 2, \dots, d\}$.

Let S be the set of all scanlines in $D(G)$.

According the above definition, clearly, $\langle S, \prec \rangle$ is a poset. Theorem 2 specifies that $\langle S, \prec \rangle$ is also a lattice, as can be easily verified based on Definitions 3 and 4.

Theorem 2. The poset $\langle S, \prec \rangle$ is a lattice. If $s_x, s_y \in \langle S, \prec \rangle$, then the meet $s_x \wedge s_y = (s^1, s^2, \dots, s^d)$ with $s^i = \min\{s_x^i, s_y^i\}$ and the join $s_x \vee s_y = (s^1, s^2, \dots, s^d)$ with $s^i = \max\{s_x^i, s_y^i\}$, for all $i \in \{1, 2, \dots, d\}$.

3 algorithm for computing K -terminal reliability of d -trapezoid graphs

Given a d -trapezoid graph $G=(V,E)$, a set K of target vertices, and the failure probability of each non-target vertex, the K -terminal reliability is defined as the probability that all target vertices can be connected to each other. If v is a target vertex in G , such that $v \in K$, then Q_v is said to be a target d -trapezoid in $D(G)$. Let each non-target Q_v fail randomly with probability q_v that equals that

of vertex v . Then, the K -terminal reliability problem in G can equivalently be expressed as the problem in which all target d -trapezoids can be connected to each other in $D(G)$.

Definition 5. Given a d -trapezoid graph $G=(V, E)$ and a set K of target vertices, the *leftmost target polyline* $L_K=(l_K^1, l_K^2, \dots, l_K^d)$ in $D(G)$ is defined by $l_K^i = \min_{v \in K} \{l_v^i\}$ and the *rightmost target polyline* $R_K=(r_K^1, r_K^2, \dots, r_K^d)$ in $D(G)$ is defined by $r_K^i = \max_{v \in K} \{r_v^i\}$, for all $i \in \{1, 2, \dots, d\}$.

Obviously, if and only if L_K and R_K are connected via all working d -trapezoids in $D(G)$, then all target d -trapezoids in $D(G)$ are connected to each other. Consequently, the K -terminal reliability of a d -trapezoid graph G is then equivalent to the probability that L_K and R_K are connected in $D(G)$. Clearly, L_K and R_K are disconnected if and only if at least one scanlines between L_K and R_K exist such that all d -trapezoids of $Q(s)$ fail. Let $S(L_K, R_K)$ denote the set of all scanlines between L_K and R_K . According to the definition of scanline, $S(L_K, R_K)$ can be expressed as $S(L_K, R_K) = \{(s^1, s^2, \dots, s^d) | s^i \in \{l_K^i + \frac{1}{2}, l_K^i + \frac{3}{2}, \dots, r_K^i - \frac{1}{2}\} \text{ for all } i \in \{1, 2, \dots, d\}\}$. The K -terminal reliability of a d -trapezoid graph is given by

$$1 - \Pr \left[\bigcup_{s \in S(L_K, R_K)} \{\text{all } Q_v \in Q(s) \text{ fail}\} \right]. \quad (1)$$

A dynamic programming approach for computing Eq. (1) is proposed. To illustrate this dynamic programming approach, for scanline s , the event $E(s) \equiv \{\text{at least one } Q_v \in Q(s) \text{ works}\}$ is defined.

Two properties of event $E(s)$ are given as follows.

Property 1. $E(s_x \wedge s_y) \setminus E(s_y)$ implies $E(s_x) \setminus E(s_y)$.

Proof. Since $s_x \wedge s_y = (s^1, s^2, \dots, s^d)$ with $s^i = \min\{s_x^i, s_y^i\}$, $Q_v \in Q(s_x \wedge s_y)$ implies $Q_v \in Q(s_x)$ or

$Q_v \in Q(s_y)$. Thus, $Q(s_x \wedge s_y) \subseteq Q(s_x) \cup Q(s_y)$ and $Q(s_x \wedge s_y) \setminus Q(s_y) \subseteq (Q(s_x) \cup Q(s_y)) \setminus Q(s_y) \equiv Q(s_x) \setminus Q(s_y)$. Therefore, $E(s_x \wedge s_y) \setminus E(s_y)$ implies $E(s_x) \setminus E(s_y)$.

Property 2. If $s_x \prec s_y \prec s_z$, then $(E(s_x) \setminus E(s_z)) \setminus E(s_y) = E(s_x) \setminus E(s_y)$.

Proof. Clearly, for $s_x \prec s_y \prec s_z$, if $Q_v \in Q(s_x)$ and $Q_v \in Q(s_z)$, then $Q_v \in Q(s_y)$. Thus, $(E(s_x) \setminus E(s_z)) \setminus E(s_y) = E(s_x) \setminus E(s_y)$.

Now, for scanline s_k , the event $R(s_k) \equiv \{\text{there is an operating path to all left scanlines of } s_k, \text{ but not to } s_k\}$ is defined.

Theorem 3 provides a dynamic programming approach for computing the probability that event $R(s_k)$ occurs. Throughout this paper, $\Pr[A]$ denotes the probability that event A occurs and $\overline{A}$ denotes the complement of event A .

Theorem 3.

$$\Pr[R(s_k)] = 1 - \sum_{s_j \prec s_k} \left(\Pr[R(s_j)] \cdot \prod_{Q_v \in Q(s_j) \setminus Q(s_k)} q_v \right).$$

Proof. Consider the complement of $R(s_k)$ and apply the sum-of-disjoint-products method.

$$\begin{aligned} \overline{R(s_k)} &= \bigcup_{s_j \prec s_k} \overline{E(s_j) \setminus E(s_k)} \\ &= \hat{\mathbf{a}} \left(\bigcap_{s_j \prec s_k} (E(s_j) \setminus E(s_k)) \cap \overline{E(s_j) \setminus E(s_k)} \right) \\ &\quad (\text{where } \hat{\mathbf{a}} \text{ denotes disjoint union}). \end{aligned} \quad (2)$$

Without loss of generality, the scanlines of $\{s_j | s_j \prec s_k\}$ in Eq. (2) are reordered by the topological ordering $<$. Accordingly, for scanlines s_i and s_j , $s_i \prec s_j$ implies $i < j$, but not vice versa. Assume s_h is the scanline such that $h < j$, but $s_h \not\prec s_j$. Consider two product terms for s_h and $s_h \wedge s_j$ in Eq.(2).

$$E_1: (E(s_h) \setminus E(s_k)) \cap \overline{(E(s_j) \setminus E(s_k))} \text{ and}$$

$$E_2: (E(s_h \wedge s_j) \setminus E(s_k)) \cap \overline{(E(s_j) \setminus E(s_k))}.$$

E_1 and E_2 are given by

$$E_1: (E(s_h) \setminus E(s_k)) \setminus E(s_j) =$$

$$(E(s_h) \setminus E(s_j)) \setminus E(s_k) \text{ and}$$

$$E_2: (E(s_h \wedge s_j) \setminus E(s_k)) \setminus E(s_j) =$$

$$(E(s_h \wedge s_j) \setminus E(s_j)) \setminus E(s_k).$$

By Property 1, $E(s_h \wedge s_j) \setminus E(s_j)$ implies $E(s_h) \setminus E(s_j)$, so if event E_2 occurs then E_1 occurs. Therefore, all product terms for s_h can be

eliminated such that $h < j$ and $s_h \not\prec s_j$ from Eq.(2). Thus,

$$\overline{R(s_k)} = \hat{\mathbf{a}} \left(\bigcap_{s_j \prec s_k} \left(E(s_i) \setminus E(s_k) \right) \bigcap \overline{E(s_j) \setminus E(s_k)} \right)$$

By Property 2, $(E(s_i) \setminus E(s_k)) \setminus E(s_j) = E(s_i) \setminus E(s_j)$ for $s_i \prec s_j \prec s_k$, so

$$\begin{aligned} \overline{R(s_k)} &= \hat{\mathbf{a}} \left(\bigcap_{s_j \prec s_k} \left(E(s_i) \setminus E(s_j) \right) \bigcap \overline{E(s_j) \setminus E(s_k)} \right) \\ &= \hat{\mathbf{a}} \left(R(s_j) \bigcap \overline{E(s_j) \setminus E(s_k)} \right). \end{aligned}$$

From the above result, the probability that event $R(s_k)$ occurs is obtained as

$$\Pr[R(s_k)] = 1 - \sum_{s_j \prec s_k} \left(\Pr \left[R(s_j) \bigcap \overline{E(s_j) \setminus E(s_k)} \right] \right).$$

Notably, $R(s_j)$ and $\overline{E(s_j) \setminus E(s_k)}$ are mutually independent events. Hence,

$$\begin{aligned} \Pr[R(s_k)] &= 1 - \sum_{s_j \prec s_k} \left(\Pr[R(s_j)] \cdot \Pr \left[\overline{E(s_j) \setminus E(s_k)} \right] \right) \end{aligned} \quad (3)$$

$$= 1 - \sum_{s_j \prec s_k} \left(\Pr[R(s_j)] \cdot \prod_{Q_v \in Q(s_j) \setminus Q(s_k)} q_v \right)$$

(where q_v is the failure probability of vertex v).

Suppose that the scanlines of $S(L_K, R_K)$ are reordered according to the topological ordering $s_1 < s_2 < \dots < s_m$, where m is the number of scanlines between L_K and R_K . To simplify the implementation of the algorithm, a dummy scanline s_{m+1} with $Q(s_{m+1}) = \emptyset$ is placed to the right of all scanlines of $S(L_K, R_K)$. Hence, $s_i \prec s_{m+1}$ for all $s_i \in S(L_K, R_K)$. Notably, the K -terminal reliability of a d -trapezoid graph is obtained as $\Pr[R(s_{m+1})]$ by Theorem 3. The algorithm for computing the K -terminal reliability of a d -trapezoid graph is formally described as follows.

Algorithm Compute the K -terminal reliability of a d -trapezoid graph

Input: use trapezoid diagram create trapezoid graph

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1   Step 1: /*create trapezoid diagram*/
2   Step 2: /*covert trapezoid diagram to trapezoid graph*/
3   Step 3: /* compute  $S(L_K, R_K)$  and sort scanlines of  $S(L_K, R_K)$  into topological order*/
4            $L_K \leftarrow$  the leftmost target polyline in  $D(G)$ ;
5            $R_K \leftarrow$  the rightmost target polyline in  $D(G)$ ;
6            $S(L_K, R_K) \leftarrow \{ (s^1, s^2, \dots, s^d) \mid s^i \in [l_K^i + \frac{1}{2}, l_K^i + \frac{3}{2}, \dots, r_K^i - \frac{1}{2}] \text{ for } i \in \{1, 2, \dots, d\} \}$ ;
7            $\{s_1, s_2, \dots, s_m\} \leftarrow$  topologically sort all scanlines of  $S(L_K, R_K)$ ;
8   Step 4: /* compute  $Q(s_i)$ ,  $1 \leq i \leq m$  */
9           for  $i \leftarrow 1$  to  $m$  do
10               $Q(s_i) \leftarrow \emptyset$ ;
11              for  $v \leftarrow 1$  to  $n$  do
12                 if  $(Q_v \cap s_i \neq \emptyset)$  then  $Q(s_i) \leftarrow Q(s_i) \cup \{Q_v\}$ ;
13              end-for
14           end-for
15   Step 5: /* compute  $\Pr[R(s_k)]$ ,  $1 \leq k \leq m+1$  */
16            $Q(s_{m+1}) \leftarrow \emptyset$ ;
17           for  $k \leftarrow 1$  to  $m+1$  do
18
19               $\Pr[R(s_k)] \leftarrow 1 - \sum_{s_j \prec s_k} \left( \Pr[R(s_j)] \cdot \prod_{Q_v \in Q(s_j) \setminus Q(s_k)} q_v \right);$ 
19           end-for
20           return  $\Pr[R(s_{m+1})]$ ;

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output: K -Terminal reliability of trapezoid graphs

end-Algorithm

Theorem 4. There is an $O(n^{2d+1})$ -time algorithm that computes the K -terminal reliability of a d -trapezoid graph G , which is given together with a d -trapezoid diagram $D(G)$.

Proof. If the sets $Q(s_i)$ for $i=1\dots m$, are implemented using m bit-vectors, then adding Q_v into $Q(s_i)$ takes $O(1)$ time; checking whether $Q_v \in Q(s_i)$ also takes $O(1)$ time, and computing the

difference between $Q(s_j)$ and $Q(s_k)$ takes $O(n)$ time. Without loss of generality, d is assumed in the following to be much less than n . It is easily to verify that the number of scanlines, m , between L_K and R_K is at most $(2n-1)^d$. Step 1 takes $O(m^2) = O(n^{2d})$ time to sort the scanlines of $S(L_K, R_K)$ in a topological ordering. In Step 2, checking whether $Q_v \cap s_i \neq \emptyset$ takes $O(d)$ time. Thus, Step 2 takes $O(m \cdot n \cdot d) = O(n^{d+1})$ time. In Step 3, checking whether $s_j \prec s_k$ takes $O(d)$ time and computing $\prod_{Q_v \in Q(s_j) \cap Q(s_k)} q_v$ takes $O(n)$ time. Hence, Step 3 takes

$O(m^2(d+n)) = O(n^{2d+1})$ time. Therefore, the total algorithm can be implemented in $O(n^{2d+1})$ time.

4 Conclusion

On this paper, we propose an extension interval graphs algorithm for k -terminal reliability. We only need an $O(n^{2d+1})$ -times for computing the K -terminal reliability of d -trapezoid graphs. In the future, we can study superclasses of trapezoid graphs reliability or merging the existing algorithms for trapezoid graphs and interval graphs into one.

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